MILITARY SPECIFICATION

SIGHT ASSEMBLY: 10527236

1. SCOPE

1.1 This specification covers one type of sight assembly consisting of a gimbal assembly, an electronic control amplifier, a gimbal positioning control, and a telescope. The sight assembly is employed in a UH-lB Helicopter enabling the gunner to acquire a target and-view a missile-target complex throughout the missile flight (see 6.1).

2* APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

FSC 1270

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DRAWINGS

U.S. Army, Frankford Arsenal

F10527236 Sight Assembly: 10527236

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3, REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples by the contractor for testing shall be as specified in the contract.

3.2 Fabrication.- The sight assembly shall be fabricated in accordance with Drawing F10527236 and drawings, specifications and standards pertaining thereto.

3.2.1 Treatment, moisture-and-fungus-resistant.- The sight assembly shall be treated for moisture and fungus resistance in accordance with MIL-T-152.

3.3 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Part identification and marking
d. Workmanship
e. Soldering electrical connections

3.4 Environmental.

3.4.1 Storage temperature.- The sight assembly shall meet the performance requirements of 3.5 at 60° to 90° Fahrenheit (F) after being subjected to ambient temperatures at -80° and at +160°F.

3.4.2 Operating temperature.- The sight assembly shall meet the performance requirements of 3.5.1 and 3.5.4 while exposed to temperatures at -40° and at +150°F.

3.4.3 Altitude.- The sight assembly shall meet the performance requirements of 3.5.1 at an altitude of 30,000 feet above sea level. - . .

3.4.4 Shock.- The sight assembly shall withstand impact shocks up to 15 G's; each shock impulse having a time duration of 11 ± 1 milliseconds.

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Subsequently, the sight assembly shall meet the performance requirements of 3.5 and retain a static balance.

3.4.5 Vibration.- The sight assembly shall meet the performance requirements of 3.5 after being subjected to vibrations which comply with curve III, MIL-E-5400. The inputs will be reduced at the lateral, fore and aft, and vertical resonant frequencies of the sight so as not to exceed 5 G's at the monocular. The sight shall retain a static balance.

3.5 Performance.

3.5.1 Stability.- The degree of stability of the servo loops as determined from a step rate response test, shall conform to the following system dynamic characteristics: The time response of each charnel from time of step-rate application to the peak of the first overshoot shall be .09 second or less. The time required for each stabilization servo transient response to settle to within 5% of its final value shall be 0.4 second. The peak overshoot shall not exceed 60% of its total excursion between time zero and its final value.

3.5.2 Elevation gyro.- The sensitive axis of the elevation gyro shall be parallel to the elevation gimbal axis within ± 1.0°.

3.5.3 Azimuth gyro.- The sensitive axis of the azimuth gyro shall be perpendicular to the sensitive axis of the elevation gyro within ± 1.0°.

3.5.4 Gimbal friction torque.- Gimbal friction shall be less than .025 pound-foot (lb.-ft.) for elevation and less than 0.1 lb.-ft. for azimuth at a room temperature of 70°F and at +150°F. Gimbal friction shall be less than .30 lb.-ft. for elevation and less than ●23 lb.-ft. for azimuth at a temperature of -40°F.

3.5.5 Natural frequency and damPin~.- The sight assembly shall have a natural frequency of at least 33 cps with a damping factor of no greater than 24.

3.5.6 Torque sensitivity.- The elevation torque motor shall develop a minimum torque of 30 ounce inches (oz. in.) per ampere of driving current. The azimuth torque motor **shall develop a minimum torque of .25 pound-foot (lb.-ft.) per ampere of driving current.**

3.5.7 Alignment of ring sight and monocular.- The polariod ring sight shall be foresighted with the monocular so that all of the visible area inside the inner ring of the ring sight will also be visible through the monocular.

3.5.8 Synchro cage.- Operation of the synchro cage switch to the **"on"** position shall cause the azimuth gimbal to rotate to within ± 2.5 degrees of the midpoint of gimbal **maximum** displacement.

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3.5.9 Trim potentiometers. - Adjustment of the azimuth and elevation trim potentiometers shall remove any drifting on the azimuth and elevation axes. The dials on the potentiometer should read in the range between "4" and "6" when the drifting is stopped. Counterclockwise rotation of the control knobs shall cause the monocular field of view to move right for the azimuth knob and up for the elevation knob.

3.6 Radio interference suppression.- Interference control requirements of the equipment shall conform to MIL-I-6181.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4 . 2 ~ . - When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 PreProduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial Production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 **First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the-contract clauses applicable to first article testing.**

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4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP'S the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 First ***article tests***.- Unless otherwise specified, the requirements and tests of Table I shall be conducted on first article tests only. In the event that the processes, materials or other changes in design or manufacture occur subsequent to first article approval, the first items so produced shall be submitted to the Government in accordance with 4.2 for approval prior to resumption of production.

TABLE I

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Altitude	3.4.3	4.6.2.3
302. Shock	3.4.4	4.6.2.4
303. Vibration	3.4.5	4.6.2.5
304. Radio interference suppression	3.6	4.6.4

4.3.2.3 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables 11 and 111 herein. The tabulated classification of defects in Tables 11 and 111 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE II - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.5 Percent Defective		
101. Stability	3.5.1	**4.6.3.1**
102. Elevation gyro	3.5.2	**4.6.3.2**
103. Azimuth gyro	3.5.3	**4.6.3.3**
104. Gimbal friction torque	3.5.4	**4.6.3.4**
105. Natural frequency and damping	3.5.5	**4.6.3.5**
106. **Torque sensitivity**	3.5.6	**4.6.3,6**
107. Alignment of ring sight and monocular	3.5.7	**4.6.3.7**
108. Synchro cage	3.5.8	**4.6.3.8**
109. Trim potentiometers	3.5.9	**4.6.3.9**
110. Moisture and fungus resistance	3.2.1	**4.6.1**
MINOR: NONE		

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- Three sights shall be selected at random as a special sampling from each 100 produced or from each months production whichever occurs first. The sample shall meet the requirements and tests in Table III.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401.	**Fabrication**	**3.2**	**Applicable drawing**
402.	**General specification**	3.3	visual MIL-F-13926 - ViSual
403.	**Temperature (storage)**	**3.4.1**	**4.6.2.1**
404.	**Temperature (operating)**	**3.4.2**	**4.6.2.2**

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SOAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Treatment, moisture and fungus-resistant.- The sight assembly shall be tested in accordance with MIL-T-152 to insure compliance with 3.2.1.

4.6.2 Environmental.

4.6.2.1 storage temperature.- The sight assembly, while in a non-operating condition, shall be subjected to the Temperature Cycling Test, Procedure II of MIL-F-13926 using Conditions A and H. Upon stabilization at ambient room temperature, the sight assembly shall meet the performance requirements specified in 3.5.

4.6.2.2 Operating temperature.- The sight assembly shall be subjected to the Temperature Cycling Test, Procedure I of MIL-F-13926 using Conditions C and G. During temperature stabilization at Conditions C, G and room ambient, the sight assembly shall meet the performance requirements specified in 3.5.

4.6.2.3 Altitude test.- The sight assembly shall be placed within a test chamber and the absolute internal pressure reduced to 8.88 inches of mercury (corresponding to an altitude of 30,000 feet above sea level). After sufficient time has elapsed to allow all entrapped air to escape,

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the sight assembly shall meet the requirements specified in 3.5.

4.6.2.4 Shock test.- The sight assembly shall be tested in accordance with MIL-F-13926, Procedure II. The sight assembly shall not suffer damage or subsequently fail to provide the performance specified in 3.5.

4.6.2.5 Vibration test.- With the sight assembly mounted and operating at room ambient conditions, the table shall be vibrated along a horizontal direction with the frequency varying between 5 and 500 cps. For frequencies between 5 and 20 cps, the double amplitude shall be 0.1 inch. For frequencies between 20 and 33 cps, the amplitude shall be varied so as to maintain a 2 G acceleration. For frequencies between 33 and 52 cps, the double amplitude shall be 0.036 inch. For frequencies between 52 and 500 cps, the amplitude shall be varied so as to maintain a 5 G acceleration. The rate of frequency change shall be logarithmic and shall be such that a complete cycle (5-500-5) will consume approximately 15 minutes. The test shall continue for four 15-minute cycles for a total of 1 hour. (when provisions for logarithmic cycling is not available, a linear rate of frequency change may be used.) The cycling test shall be repeated with the axis of vibration changed 90 degrees in the horizontal direction and then with the axis of vibration changed to vertical. At the completion of each direction of vibration, the sight assembly shall be closely inspected for any evidence of mechanical failure, loose or broken parts. At end of test, the sight assembly shall meet the performance requirements of 3.5.

4.6.3 Performance.

4.6.3.1 Stability.- Attach a recorder to the output of the torquers on the pitch and azimuth axes. With the system properly connected and in an operating condition, the degree of stability shall be within the limits of 3.5.1 as shown on the tape of the recorder.

4.6.3.2 Elevation gyro.- Simultaneously apply the pick-off signal of the azimuth gyro and the pick-off signal of the elevation gyro to a calibrated dual channel demodulator and recorder. Adjust the gains on both channels until equal amplitudes are obtained. With the elevation gimbal locked, the azimuth gimbal shall be slewed to produce an azimuth signal which is at least 50% of the maximum rated gyro output. Compare the two recorded signals to verify that the elevation gyro signal is less than 0.9% of the azimuth gyro output. Test results within the indicated percentage shall be the equivalent of, and in compliance with 3.5.2.

4.6.3.3 Azimuth gyro.- Perform the same test as 4.6.3.2, except lock the azimuth gimbal and slew the elevation gimbal. Verify test results for compliance with 3.5.3.

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4.6.3.4 Gimbal friction torque.- Apply a torquer signal to the elevation torquer when the elevation gimbal is approximately in the 0°, +10°, and -10° positions. Visually observe the gimbal to determine the minimum current required to produce motion. The average current of the three positions shall not exceed 110 milliamperes at +70°F and +150°F nor exceed 1500 milliamperes at -40°F. Apply a torquer signal to the azimuth torquer when the azimuth gimbal is approximately in the 0°, +30° , -30° positions. Visually observe the gimbal to determine the minimum current required to produce motion. The average current of the three positions shall not exceed 333 milliamperes at +70°F and +150°F nor exceed 750 milliamperes at -40°F. Test results within the indicated values shall be equivalent of, and in compliance with 3.5.4.

4.6.3.5 Natural frequency and damping.- The sight assembly is to be mounted in a holding fixture. With an accelerometer mounted on the yoke of the sight assembly, a force impulse shall be imparted along the lateral axis and the output of the accelerometer and amplifier recorded on a chart. The values for the natural frequency and damping shall be in compliance with 3.5.5.

4.6.3.6 Torque sensitivity.- Apply a current of one ampere to the " elevation torquer。 Using a pull scale, hook the end over the tip of the counterweight nut that attaches the monocular to the gimbal (a radius of 4.12 inches). Measure the tangential force. The scale shall read a minimum of 30 ounce inches (oz. in.). Apply a current of two amperes to the azimuth torquer. Attach the pull scale'to the hole in the bottom of the azimuth counterweight (a radius of 13 inches). Measure the tangential force. The scale shall read a minimum of .25 pound-foot (lb.-ft.). Test results within the indicated values shall be equivalent of, and in compliance with 3.5.6.

4.6.3.7 Alignment of ring sight and monocular.- The monocular shall be aimed at some point at least 1000 feet away. The polarized ring sight will be adjusted until the same point is in the center of the inside ring. After adjustment, check all of the objects in the inner ring and see if they are visible through the monocular. This ***test*** will satisfy the requirements of 3.5.7.

4,6.3.8 Synchro cage.- Operate the synchro cage switch to the "ON' position and check for compliance with the requirements of 3.5.8.

4.6.3.9 Trim potentiometer.- With the sighting system turned on and operating, adjust the azimuth and elevation trim potentiometers and check for compliance with the requirements of 3.5.9.

4.6.4 Radio interference suppression test.- The equipment shall be tested in accordance with MIL-I-6181 to insure conformance with 3.6.

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5. PREPARATION FOR DELIVERY

(NOT APPLICABLE)

6. NOTES

6.1 Intended use. - The Sight Assembly: 10527236, part of the Anti-oscillation Sighting System: 10527235, is intended to be used in a UH-lB Helicopter. The sight assembly uses a two-degree-of-freedom gimbal arrangement permitting free rotation around the pitch and yaw axes of the helicopter. Attached to the elevation gimbal are two rate gyros that function as inertial angular disturbance sensors. These sensors supply error-corrected signals to the yaw and pitch torque motors (torquers) through the electronic control amplifier to correct for apparent pre-cession.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification~
b. Applicable stock number.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No.1270-A039y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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